

# A counting bound for hidden–zero survivors: a crossing costs one extra chord

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## Abstract

We formalize the following idea: surviving the hidden–zero (“ $K$ –zero”) conditions forces a term’s chord arrangement to be *fully triangulated*, and a fully triangulated arrangement with  $p$  crossings needs exactly  $m = n - 3 + p$  chords. Since a  $\text{Tr } \phi^3$  term carries exactly  $m = n - 3$  chords, survival forces  $p = 0$ : only triangulations survive. The Euler count is rigorous; the whole conjecture reduces to one local statement, the *Face Lemma*, that the  $K$ –zeros triangulate every face.

## 1 The arrangement and the Euler count

Fix a convex  $n$ –gon, vertices  $1, \dots, n$ . A term is a set  $M$  of  $m$  chords; draw them as straight segments in general position (no three chords concurrent — the statement is combinatorial, so this is no loss). Let  $p$  be the number of crossing pairs. The induced planar subdivision has

$$V = n + p, \quad E = n + (m + 2p), \quad F = E - V + 2 = m + p + 2,$$

where  $E$  counts the  $n$  boundary edges plus the chord segments: a chord crossed  $c$  times contributes  $c + 1$  segments, and  $\sum_{\text{chords}} c = 2p$ , giving  $m + 2p$  segments. Hence the number of *bounded* faces is

$$F_{\text{int}} = F - 1 = m + p + 1. \tag{1}$$

**Lemma 1** (Counting). *If every bounded face of the arrangement is a triangle, then*

$$m = n - 3 + p.$$

*Proof.* Sum the face degrees. Each bounded face is a triangle (3 sides); the unbounded face is bounded by the  $n$  boundary edges. As  $\sum_f \deg(f) = 2E$ ,

$$3F_{\text{int}} + n = 2E \implies 3(m + p + 1) + n = 2(n + m + 2p),$$

using (1). This simplifies to  $3m + 3p + 3 + n = 2n + 2m + 4p$ , i.e.  $m = n - 3 + p$ . □

**Remark 1.** Checks: a triangulation has  $p = 0$ ,  $m = n - 3$ , and  $F_{\text{int}} = n - 2$  triangles. The two diagonals of a square have  $p = 1$  and  $m = 2 = n - 3 + p = 1 + 1$ , with four triangular faces. These agree with (1) and Lemma 1.

## 2 The reduction

**Theorem 1.** *Assume the Face Lemma below. Then any  $\text{Tr } \phi^3$  term that survives all  $K$ -zeros is a triangulation.*

*Proof.* A term carries exactly  $m = n - 3$  chords. By the Face Lemma its arrangement is fully triangulated, so Lemma 1 gives  $n - 3 = n - 3 + p$ , hence  $p = 0$ . A crossing-free set of  $n - 3$  chords is a triangulation.  $\square$

**Lemma 2** (Face Lemma — the remaining target). *If  $M$  survives all  $K$ -zeros, every bounded face of its arrangement is a triangle.*

**What is proved.** For a face bounded entirely by *full* chords (no crossing point on its boundary): such a face is an empty sub-polygon with  $\geq 4$  sides, i.e. a window whose interior vertices are only connected within it and whose closing chord does not extend to a  $(k+2)$ -gon. This is exactly the configuration that note 21 shows vanishes. So no such face can survive.

**What remains.** Faces having a crossing point as a corner are bounded by chord *segments*, where the note 21 statement (phrased for full chords/vertices) does not directly apply. One must show that the finer covering condition (note 25, point (C)) forces the wedges around each crossing to be filled in — the cascade “a  $(k+2)$ -gon on top of every chord, or a shoot-out.” By Lemma 1 this cascade cannot terminate within the budget  $m = n - 3$  once  $p \geq 1$ , which is the geometric reason the bound must hold; turning that into a proof is the open step.

## 3 Illustration: the $n = 12$ pizza

The pizza  $\Pi = \{(1, 3), (3, 5), (5, 7), (7, 9), (9, 11), (11, 1)\} \cup \{(2, 8), (4, 10), (6, 12)\}$  has  $m = 9$  chords and  $p = 9$  crossings. To triangulate its arrangement, Lemma 1 demands  $m = n - 3 + p = 9 + 9 = 18$  chords. With only 9, the arrangement necessarily contains non-triangular faces; by the Face Lemma some  $K$ -zero kills it. This is the counting-side view of note 25’s “ $k = 2$  kills the pizza.”

## 4 Status

The Euler count (Lemma 1) and the reduction (Theorem 1) are rigorous. The conjecture “only triangulations survive” is now equivalent to the single local statement of Lemma 2: *the  $K$ -zeros triangulate every face*. The only unproved piece is the behaviour of faces incident to a crossing point, where the note 25 covering condition must be applied carefully.